

Nimbus — How It Works

Version: 1.3 **Author:** Sujal Kumar Singh **Last updated:** 2026-08-19

This is the plain-English version. Read this to understand the system.

Doc	For
OVERVIEW.md (this one)	Understanding what we are building and why
ARCHITECTURE.md	Every part of the system drawn as a diagram
HLD.md	Exact table columns, endpoints, and steps. The build reference.

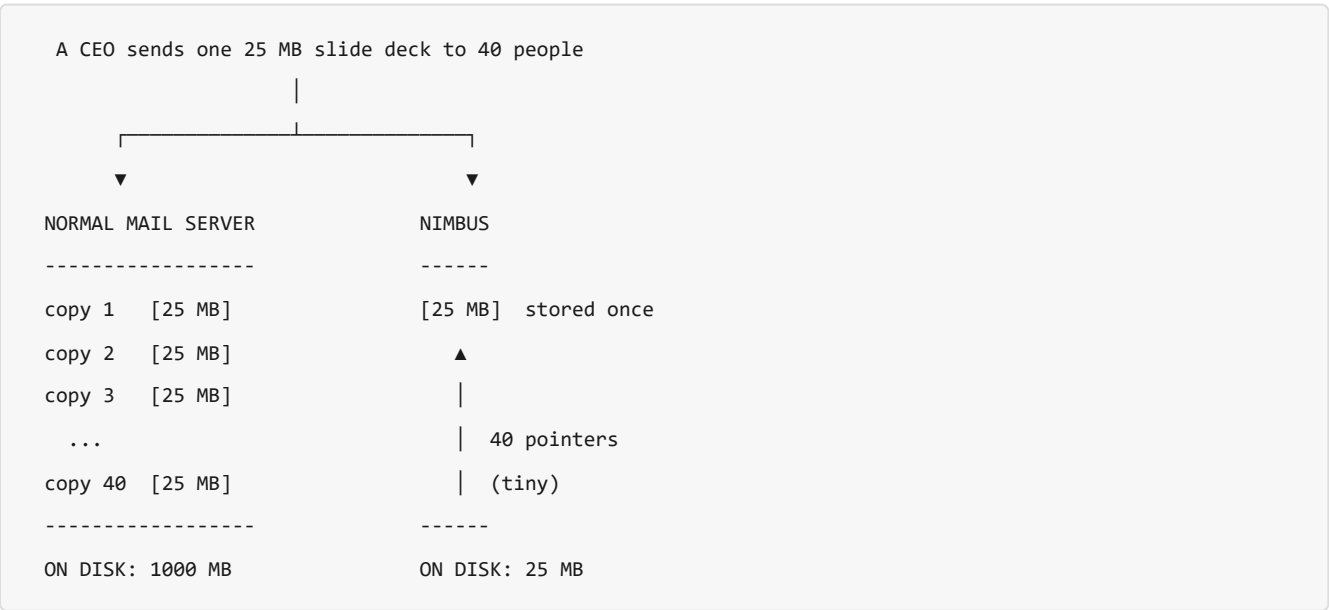
1. The whole thing in one line

A mail server that stores every attachment once, no matter how many people receive it.

2. The problem

Business email costs about **\$2 per mailbox per month**. At that price, the biggest cost is storage. So storage is where you win or lose money.

Here is what goes wrong with a normal mail server:



Same email. Same 40 people. **40x less disk.**

Now imagine 10 million mailboxes instead of 40. That gap decides whether the business makes money or not.

3. The trick

We never ask *"what is this file called?"*. We ask *"what is this file made of?"*.

any bytes → SHA-256 → 3a7f9c2e...d41 (64 characters)

same bytes → same fingerprint → we have it, store nothing

one bit differs → totally different → new file, store it

That fingerprint is the file's name in our storage. This is called **content-addressed storage**.

Three things fall out of it for free:

- **No duplicates.** Same fingerprint means we already have those bytes.
- **Writing twice is safe.** Two people uploading the same file at the same moment both write to the same place. No conflict, no lock needed.
- **Damage is detectable.** If the bytes ever change, the fingerprint stops matching.

How the data actually sits

ONE message row

id	501
subject	Q3..
from	ceo

ONE blob row

hash	3a7f9c..
size	25 MB
refcount	40

fan-out

sujal	ravi	asha	...
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40 mailbox_message rows
each one is about 40 bytes

the bytes
live once

- **1 message row** — the email itself, stored once
- **1 blob row** — the attachment, stored once
- **40 tiny rows** — one per person, so each person sees it in their own inbox

Everyone gets their own copy in their inbox. Nobody gets their own copy on disk.

4. Goals — what "done" means

#	Goal	Done when
1	Accept real email	A real mail server can send us mail and we keep it
2	Store each file once	Same attachment to 40 people = 1 copy on disk
3	Deliver to many, copy nothing	40 recipients = 40 small rows, 0 extra bytes
4	Search properly	<code>from:boss</code> <code>has:attachment</code> <code>invoice</code> answers in under 100 ms
5	Show what we saved	Dashboard: "they think 1 GB, we stored 25 MB"
6	Free space safely	Deleting the last copy frees the disk, and never too early
7	Let a reseller sign customers up	One API call creates a domain and its mailboxes

5. What we are NOT building

This matters as much as the goals. It stops the project growing forever.

- **We do not send email.** No outbound. Nimbus only receives.
- No Send Later, no Unsend, no read receipts — all of those need sending.
- No calendar, contacts, signatures, templates.
- No IMAP or POP3. Browser only.
- No spam filtering beyond basic sender checks.
- No mobile apps.

Why receive-only? Because dedup only helps when you *store*. When you send a 25 MB file to 40 people, SMTP makes you transmit all 1000 MB anyway. There is no trick for that. Storage is where the saving is real, so that is where we work.

5.1 There is no admin screen, and that is on purpose

Nimbus has three kinds of user, and only the last one has a screen:

Who	What they do	How they do it
Us	Create a customer organization	Run a script on the server
The organization	Create their domain and their staff's mailboxes	Call the API with their key
A person	Read their mail	The web app

So signing up a new customer needs somebody with access to the server. That sounds like a missing feature and mostly it is not one. Business email is sold, not self-served — there is a contract and an invoice before there is an account, so Mailgun and Postmark make you talk to a salesperson too. **The API is the customer product; the script is the admin product.** What was genuinely wrong is that this was never written down as a decision, so it read as an oversight. Now it is written down. (HLD.md §9.8.)

What *is* missing and is being built: the organization can create things but can never remove them. No delete-a-mailbox, no delete-a-domain. In a system whose whole point is reclaiming storage, the operation that releases storage has no button. The database already handles it correctly — only the API is absent. That is block L2.

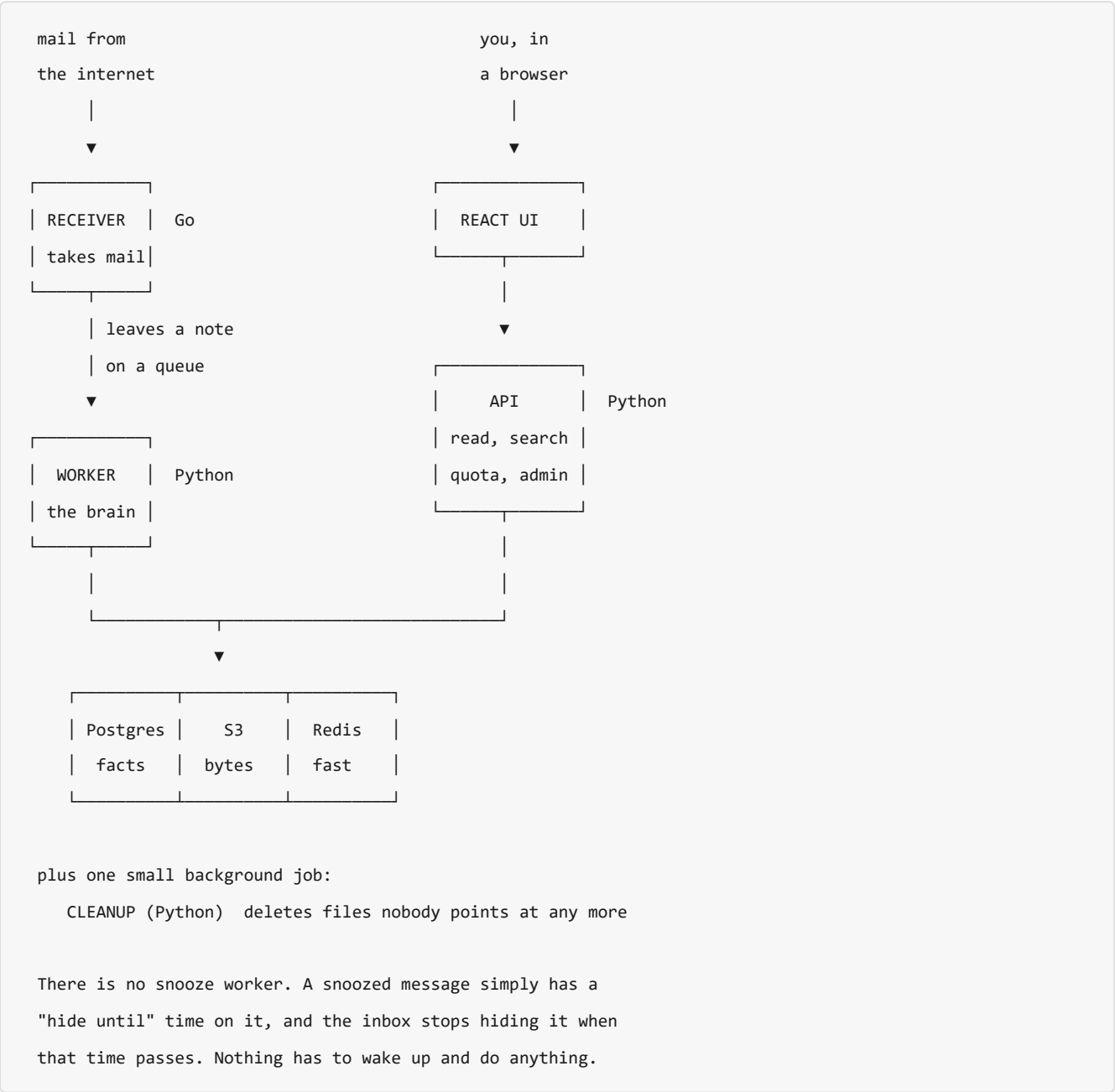
5.2 We never checked that a domain belongs to the customer

Anyone with a valid API key could claim `google.com`, because we only ever checked that no one else had claimed it first. There is a column in the database called `verified` that was supposed to prevent exactly this. Nothing has ever set it.

Right now that is contained — API keys are handed out by us, one at a time. It stops being contained the moment the system is on the real internet with a real address, because then "claimed a domain it does not own" means "is accepting somebody else's mail". So it gets fixed before the deploy, not after.

The fix is the standard one. We give the customer a random-looking string. They have to put it into their domain's DNS settings — which only the real owner can do. We look it up. If it matches, the domain is verified and its addresses go live. If not, mail to that domain keeps bouncing. Google Workspace and Let's Encrypt prove ownership the same way. (HLD.md §9.6a.)

6. The parts



Part	Language	Its one job	Why that language
Receiver	Go	Take mail off the wire, put it in S3	Many slow connections at once, flat memory. Capped at 100 so a flood cannot exhaust the box.
Worker	Python	Split, fingerprint, dedup, deliver, index	The thinking part. Nothing here is urgent.
API	Python	Everything the browser asks for	Most of the code. Speed of writing matters most.
Cleanup	Python	Delete unused files	Runs in the background. Slow is fine.

Where things are kept

Store	Holds	Why there
Postgres	Who owns what, who got which email, all counts	Needs to be exactly right. Source of truth.
S3	The actual bytes of files and emails	Files must never go in a database. Cheap, endless.
Redis	Which addresses exist, so the receiver can answer RCPT TO in under a millisecond	It is a cache — rebuilt from Postgres at startup, so nothing lives here that we could not rebuild
Kafka	The queue of "new mail arrived" notes	If a traffic spike hits, mail waits instead of being lost

7. What happens when an email arrives

```
1 Another mail server connects to us
  |
2 It says "I have mail for sujal@acme.com"
  We check right now: does that address exist?
  no  → we reply "550 No such user"  <-- our ONLY chance to say no
  yes → carry on
  |
3 The email arrives. We write it straight into S3
  as it flows in. We never hold the whole thing in memory.
  |
4 We leave a note on the queue: "new mail, here is where
  it is". Then we hang up. The sender is done.
  |
5 A worker picks up the note and does the real work:
  - split the email into text and attachments
  - fingerprint each attachment
  - do we already have it?
    yes → store nothing at all
    no  → upload it
  - work out who actually gets it (see section 8)
  - write one small row per person
  - add it to the search index
```

Why step 2 matters so much. Step 2 is the last moment we can refuse a message. After that the sender has hung up and gone. By step 5 there is nobody left to talk to. So every check that could reject mail has to happen at step 2, not later.

Why step 3 matters. A 25 MB email must not use 25 MB of memory. We copy it through to S3 in small pieces as it arrives. 1 email or 1000 at once, memory stays flat.

Why step 4 matters. Once the note is on the queue, the mail is safe. If the worker crashes, the note is still there and gets picked up again. Nothing is lost.

One catch with step 5. The queue can hand us the same note twice — that is normal and expected. If we just processed it again, the same email would appear in someone's inbox twice and our counts would go wrong. So the worker writes down "I have done this one" first, in the same database transaction as everything else. A repeat instantly collides with that record and stops.

8. Who actually gets the email

An address is not always a person. We check in this order and stop at the first match.

```
mail for sales@acme.com
  |
1 Is it a nickname for someone? → send to that person
  | no
2 Is it a shared inbox?         → send to every member
  | no
3 Does it forward somewhere?   → forward it on
  | no
4 Does the domain catch everything? → drop it in that inbox
  | no
5 Nothing matched               → log it and drop it
```

Step 5 is not a bounce, and it cannot be. We already said yes to this address back in section 7 step 2. The connection is gone. Sending a real bounce email would mean sending mail, which section 5 says we do not do. In practice step 5 only happens if someone deleted the mailbox in the few seconds in between.

9. Opening an email with a big attachment

The file might be 25 MB. We do not load 25 MB to hand it over.

blob 3a7f9c.. is stored as pieces:



We read piece 1, send it, forget it.

Then piece 2. Then piece 3.

Memory used: one piece at a time. Never the whole file.

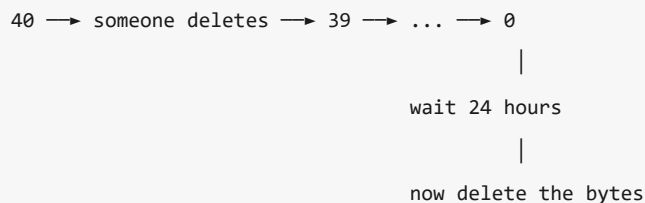
This also means a download can resume if the connection drops.

10. Deleting things without losing data

This is the hardest part of the whole project.

Every stored file counts how many people still point at it. That count is the **refcount**.

refcount = how many inboxes still hold this file



Why wait 24 hours instead of deleting at zero?

Mailbox A: deletes its copy refcount goes 1 → 0
Mailbox B: is uploading the SAME (at that exact moment)
 file right now

Delete immediately: we wipe the bytes B is about to point at.
 B now has a broken attachment.

Wait 24 hours: B's upload pushes the count back to 1
 long before the sweep runs. Nothing breaks.

The delay costs us a little disk space for a day. It saves us from needing a global lock across the entire system, which would be far more expensive and far more fragile.

11. What the user sees vs what we store

	Meaning	Example
Logical	What the user thinks they used	1 GB
Physical	What we actually put on disk	25 MB

We bill logical. Three reasons:

- The user's mental model is "my mailbox holds 1 GB". Billing anything else is confusing.
- The saving is our profit margin, not a discount we give away.
- Physical is what we track to prove the engine works.

The dashboard shows both side by side. That contrast is the demo.

12. The hard problems, in plain words

Problem	What we do
Two people get the same file at the same second	Fingerprints make writing twice harmless. A short Redis lock stops us uploading it twice for nothing.
When is it safe to delete?	Count the pointers. Wait 24 hours at zero.
Encryption breaks dedup	The same file encrypted with two different keys looks like two different files. For now we use one key for storage, so dedup still works. Noted trade-off.
Dedup can leak information across customers	If an upload is suspiciously instant, you learn someone else already has that file. Real published attack on Dropbox. We fix it by only deduplicating within one customer, never across.
Receiving 25 MB without using 25 MB of RAM	Stream it through in pieces. Never hold the whole thing.
4 MB pieces — is that the right size?	For most email, attachments are smaller than 4 MB anyway, so this rarely matters. It matters for big files. Smarter splitting is an upgrade later if we need it.

13. How we know it worked

All six have been measured.

What we measure	Target	What we got
Disk saved on realistic mail	more than 60%	68.8% on 10,000 emails
Mail accepted per minute	more than 500, kept up	500, kept up — never more than 6 emails behind
Memory while receiving 25 MB files	flat — does not grow with file size	212 MB of mail through 84 MB of memory
Search on a 100,000-message mailbox	under 100 ms for 95% of searches	5 ms for an ordinary word, 73–76 ms for a search matching nearly the whole mailbox
Snooze accuracy	exact — nothing fires, so nothing can be late	exact
Timers we can hold at once	1 million, no slowdown	nothing to slow down — they are just rows

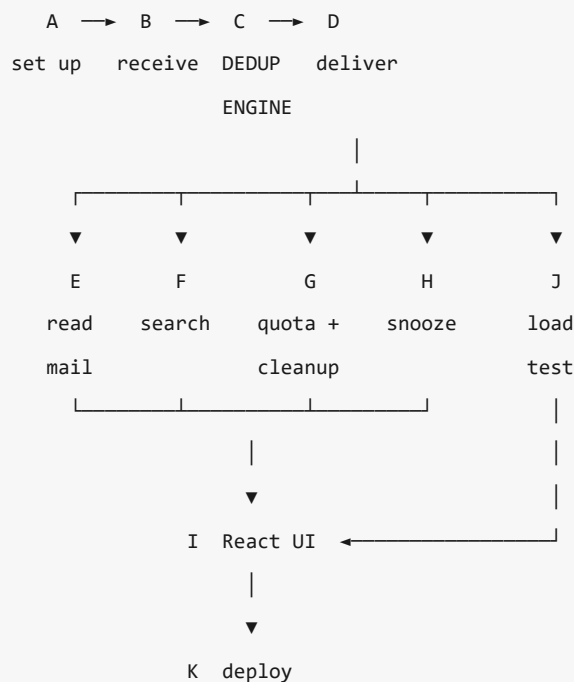
Important honesty note on the first row. That number is decided by the test emails we generate, not by the code. Random files dedup at 0%. All-identical files dedup at 100%. Neither proves anything. So we write down exactly what the test set contains and publish it next to the number — and the whole test set is reproducible from a single number, the "seed". Change the seed and the same recipe gives anything from **65.7% to 78.3%**. That swing is not noise to hide; it is the honest width of the claim.

And a second honesty note, on what "saved" counts. 68.8% is the saving on *attachments*. We also keep the original raw email for 7 days, and that copy is not deduplicated at all — forty emails carrying one slide deck keep forty full copies of it. Measured, that raw pile was **4.4 times bigger** than the deduplicated store. So the saving on the disk you actually pay for is **68.4% today, and 94.2% once the 7-day copies expire**. The bigger number is real, it just arrives a week late.

We also found the maximum. 500 a minute is what we *asked* for, and the system kept up without straining. So we asked for 6,000 a minute to see where it breaks. It stored **1,417 a minute** — nearly three times the target — and the part that ran out of road was the **worker** (the brain), not the receiver (the part taking mail off the wire). The receiver was still accepting twice what the worker could store.

The important half of that result is what did *not* happen. At three times the target the queue built up to 5,777 waiting emails — and then drained to zero, with every one of the eleven correctness checks still passing and not a single email lost. Overloading it made it slow, not wrong. That is exactly the job the queue exists to do.

14. Build order



Block	What	Notes
A	Docker, database, login, sign-up API	Done. Everything waits on this
B	The SMTP receiver	Done. 212 MB of mail through 84 MB of memory
C	The dedup engine	Done. The point of the whole project
D	Work out who gets what, write the rows	Done.
E	Read your inbox, download attachments	Done.
F	Search	Done. from:boss has:attachment invoice works
G	Quota and cleanup	Done. The sweep that actually gives the disk back
H	Snooze	Done. No worker — the mail just comes back when the time passes
I	The React screens and the savings dashboard	
J	Load test with 100,000 emails	Takes ~3.5 hours of real time. Start it early.
K	Deploy to AWS	

Two things take real-world time and cannot be rushed:

1. The load test itself runs for about 3.5 hours. Start it as soon as D works and build other things while it runs.
2. AWS blocks port 25 on new accounts and takes several days to unblock it. **File that request before writing any code.** It is free and it is slow.

15. Words you need

Word	What it means
SMTP	The language mail servers use to hand email to each other
MIME	The format inside an email that lets it carry text and files together
MX record	A DNS entry saying "mail for this domain goes to that server"
Blob	One whole attachment
Chunk	A 4 MB piece of a blob
SHA-256	Turns any bytes into a fixed 64-character fingerprint
Content-addressed	Naming a file by its fingerprint instead of its filename
Dedup	Storing identical content only once
Refcount	How many things still point at a file
Fan-out	One email creating rows in many inboxes
Idempotent	Doing it twice has the same result as doing it once
Multi-tenant	Many separate customers sharing one system safely
Stream	Move data through in small pieces instead of loading it all
Grace period	A deliberate delay before deleting, to avoid a race
p95	95 out of 100 requests were faster than this

16. Where to go next

You want	Read
Exact database columns	<code>HLD.md</code> §8
Exact API endpoints and login	<code>HLD.md</code> §10
Step-by-step flows with real detail	<code>HLD.md</code> §9
Decisions still open	<code>HLD.md</code> §16
How we work on this project	<code>../CLAUDE.md</code>